

Tesco Smart Checkout Optimisation – Business Analysis Case Study

Business Requirements Document (BRD)

By
Adithya Santhosh Nair

Date:
May 2026

1. Background

Tesco stores experience significant increases in customer traffic during peak shopping periods, including evenings, weekends, and holidays. Checkout queues become longer as customer demand exceeds available staffed checkouts.

The current staffing approach relies primarily on fixed schedules and manual management decisions, making it difficult to respond quickly to unexpected increases in customer demand.

2. Problem Statement

Inefficient staff scheduling and limited visibility of real-time customer demand result in insufficient checkout capacity during peak trading periods.

This causes:

- Increased customer waiting times
- Customer frustration
- Basket abandonment before purchase
- Reduced employee productivity
- Lower customer satisfaction
- Potential loss of revenue

3. Business Objectives

- Reduce customer waiting times during peak trading hours.
- Improve the efficiency of checkout operations.
- Optimise staff allocation based on customer demand.
- Reduce basket abandonment caused by long queues.
- Improve customer satisfaction while maintaining operational efficiency.

4. Project Scope

4.1 Purpose of the Project

The purpose of this project is to analyse Tesco's current checkout operations and define the business requirements for a smart checkout optimisation solution. The proposed solution aims to reduce customer waiting times, improve staff allocation, enhance customer satisfaction, and support operational decision-making through real-time monitoring and predictive analytics.

4.2 In Scope

The following features and activities are included within the scope of this project:

Business Analysis

- Analyse the current checkout process.
- Identify operational bottlenecks affecting checkout efficiency.
- Gather and document business requirements.
- Define functional and non-functional requirements.
- Conduct stakeholder analysis.
- Assess project risks and business impacts.

Queue Management

- Monitor real-time checkout queue lengths.
- Measure average customer waiting time.
- Monitor checkout utilisation.
- Generate queue performance reports.

Staff Allocation

- Recommend optimal checkout staffing levels.
- Generate alerts when additional checkouts should be opened.
- Support dynamic staff allocation during peak trading periods.

Dashboard & Reporting

- Develop a management dashboard displaying:
 - Queue lengths
 - Waiting times
 - Active checkouts
 - Staff allocation
 - Customer traffic
 - Key Performance Indicators (KPIs)

AI-Assisted Decision Support

- Forecast customer demand using historical and real-time operational data.
- Recommend proactive staffing adjustments.
- Generate automated operational alerts.

4.4 Project Deliverables

The project will produce the following deliverables:

- Business Requirements Document (BRD)
- Stakeholder Analysis
- Business Process Analysis (As-Is & To-Be)
- Root Cause Analysis
- Functional Requirements
- Non-Functional Requirements
- User Stories
- Product Backlog
- UML Use Case Diagram
- BPMN Process Diagram
- Risk Register
- Cost–Benefit Analysis
- Dashboard Wireframe
- Executive Presentation

4.5 Project Success Criteria

The project will be considered successful if it achieves the following objectives:

- Reduce average customer waiting time during peak trading hours.
- Improve customer satisfaction with the checkout experience.
- Optimise checkout staff allocation.
- Reduce basket abandonment caused by long queues.
- Provide store managers with real-time operational insights.
- Support faster, data-driven operational decisions.
- Improve overall checkout efficiency while maintaining service quality.

5. Stakeholder Analysis

5.1 Purpose

The purpose of the stakeholder analysis is to identify all individuals, departments, and external parties affected by the Tesco Smart Checkout Optimisation project. Understanding stakeholder needs and expectations helps ensure that the proposed solution aligns with business objectives and supports successful project delivery.

5.2 Stakeholder Register

Stakeholder	Role	Interest in Project	Influence	Engagement Strategy
Customers	End users of the checkout service	Shorter waiting times, faster checkout, better shopping experience	High	Customer feedback, satisfaction surveys
Checkout Colleagues	Operate tills and serve customers	Efficient workflow, manageable workload	High	Workshops, interviews, training sessions
Shift Leaders	Manage daily store operations	Better staff allocation and queue management	High	Regular project meetings and requirement validation
Checkout Team Manager	Oversees checkout operations	Improved checkout performance and KPI achievement	High	Weekly progress reviews and stakeholder meetings
Store Manager	Project Sponsor	Operational efficiency, increased sales, improved customer satisfaction	Very High	Decision-making meetings and project approvals
Regional Operations Manager	Oversees multiple stores	Standardised operations and performance improvements	High	Monthly progress updates

Workforce Planning Team	Creates staffing schedules	Accurate workforce planning and efficient staff utilisation	Medium	Requirements workshops
IT Department	Supports system integration	Successful implementation and system reliability	High	Technical workshops and design reviews
Data Analytics Team	Develops forecasting models	Accurate demand forecasting and reporting	High	Collaborative solution design sessions
Finance Department	Reviews project investment	Positive return on investment and cost control	Medium	Business case and cost-benefit reviews
HR Department	Training and workforce management	Staff readiness and change adoption	Medium	Training planning meetings
Compliance & GDPR Team	Ensures legal compliance	Data privacy and regulatory compliance	High	Compliance reviews and approval
AI / Software Vendor	Provides AI solution	Successful deployment and ongoing support	Medium	Technical coordination meetings
Tesco Executive Management	Strategic oversight	Business value and organisational benefits	Very High	Executive presentations and milestone reports

5.3 Stakeholder Power–Interest Matrix

This helps determine how each stakeholder should be managed.

Manage Closely (High Power, High Interest)

- Store Manager
- Checkout Team Manager
- Shift Leaders
- IT Department
- Data Analytics Team
- Tesco Executive Management

Keep Satisfied (High Power, Lower Daily Interest)

- Regional Operations Manager
- Finance Department
- Compliance & GDPR Team

Keep Informed (Lower Power, High Interest)

- Customers
- Checkout Colleagues
- Workforce Planning Team
- HR Department

Monitor (Lower Power, Lower Interest)

- AI / Software Vendor (until implementation phase)

5.4 Communication Plan

Stakeholder	Communication Method	Frequency
Store Manager	Project meetings	Weekly
Shift Leaders	Team workshops	Weekly
Checkout Colleagues	Training sessions	Bi-weekly
IT Department	Technical meetings	Weekly

Data Analytics Team	Sprint reviews	Weekly
Finance Department	Budget reviews	Monthly
Executive Management	Project status presentation	Monthly

6. Current State Analysis

6.1 Overview

Tesco stores currently manage checkout operations using a predominantly manual and reactive approach. During normal trading hours, checkout staffing is based on pre-planned shift schedules and manager experience rather than real-time customer demand.

When customer traffic increases unexpectedly, additional checkout tills are opened only after long queues become visible to store colleagues or are reported by customers. This reactive approach can lead to service delays, longer customer wait times, and inconsistent operational performance across trading periods.

Although historical sales data and workforce schedules are available, they are not fully utilised to proactively predict customer demand or optimise staff allocation during peak trading hours.

6.2 Current Checkout Process

The current checkout process follows these steps:

1. Customers select products within the store.
2. Customers join an available checkout queue.
3. Checkout colleagues process purchases manually.
4. Shift Leaders or Store Managers monitor queue lengths through observation.
5. Additional tills are opened only when queues become excessively long and staff are available.
6. Customers complete payment and leave the store.

6.3 Current Challenges

The analysis identified several operational challenges within the existing checkout process.

Staffing Challenges

- Fixed staff schedules may not align with actual customer demand.
- Limited flexibility to redeploy colleagues during busy periods.

- Additional tills are often opened reactively rather than proactively.

Operational Challenges

- Queue monitoring relies heavily on manual observation.
- No standard threshold exists for opening additional checkouts.
- Decision-making depends on individual manager experience rather than consistent data.

Customer Challenges

- Long waiting times during peak shopping hours.
- Increased customer frustration.
- Basket abandonment due to excessive queues.
- Inconsistent checkout experience across different stores and shifts.

Technology Challenges

- Limited use of predictive analytics.
- No automated queue monitoring.
- Lack of real-time operational dashboards.
- No automated staffing recommendations.

6.4 Root Causes

The primary root causes identified are:

- Staff scheduling does not consistently reflect customer demand.
- Operational decisions are largely reactive.
- Limited visibility of real-time queue performance.
- Underutilisation of available operational data.
- Lack of technology to support proactive workforce planning.

6.5 Business Impact

The current operating model has several consequences for Tesco.

Customer Impact

- Longer checkout waiting times.
- Reduced customer satisfaction.
- Increased likelihood of customers abandoning purchases.

Operational Impact

- Inefficient utilisation of checkout colleagues.
- Increased pressure on Shift Leaders during peak periods.
- Inconsistent decision-making across stores.

Financial Impact

- Potential loss of revenue from abandoned baskets.
- Higher operational costs due to reactive staffing decisions.
- Reduced productivity during peak trading periods.

6.6 Current State Summary

The current checkout management process relies on manual observation and reactive operational decisions. While experienced managers can respond to changing customer demand, the absence of real-time monitoring, predictive analytics, and structured decision support limits operational efficiency and affects the customer experience.

The findings from this analysis indicate an opportunity to improve checkout operations through better use of operational data, enhanced workforce planning, and technology-supported decision-making.

7. Future State Analysis

7.1 Overview

The proposed future state introduces a proactive and data-driven checkout management process. Instead of relying solely on manual observation and fixed staffing schedules, store managers and shift leaders will use real-time operational data and predictive analytics to make informed staffing decisions before queues become excessive.

The solution aims to improve customer experience, optimise workforce utilisation, and support operational decision-making through intelligent monitoring and automated recommendations.

7.2 Future Business Process

The proposed process includes the following activities:

1. Customer traffic is continuously monitored.
2. Historical and real-time operational data are analysed.
3. The system predicts expected customer demand.
4. Queue lengths and waiting times are monitored automatically.
5. The system recommends opening additional checkouts when predefined thresholds are reached.
6. Shift Leaders review and approve recommendations.
7. Available colleagues are reassigned where appropriate.

8. Store Managers monitor performance through operational dashboards.
9. Performance reports are generated to support continuous improvement.

7.3 Future Capabilities

Following implementation, the business will be able to:

Operational Monitoring

- Monitor queue lengths in real time.
- Monitor average customer waiting time.
- View checkout utilisation.
- Monitor staffing levels.

Decision Support

- Receive automated alerts.
- Receive AI-assisted staffing recommendations.
- Identify busy trading periods before queues become excessive.
- Allocate staff more efficiently.

Reporting

- View operational KPIs.
- Analyse daily and weekly performance.
- Identify recurring operational bottlenecks.
- Support continuous improvement initiatives.

7.4 Expected Business Benefits

Customer Benefits

- Reduced waiting times.
- Faster checkout experience.
- Improved customer satisfaction.
- Fewer abandoned purchases.

Employee Benefits

- Improved workload distribution.
- Reduced operational pressure.
- Better visibility of priorities.
- Improved decision support.

Business Benefits

- Improved operational efficiency.
- Increased revenue through reduced basket abandonment.
- Better utilisation of existing staff.
- More consistent service across stores.
- Data-driven operational planning.

7.5 Future State Success Measures

The success of the future operating model will be measured using the following Key Performance Indicators (KPIs):

KPI	Current State	Target
Average Queue Waiting Time	8 minutes	Under 3 minutes
Customer Satisfaction	78%	Above 90%
Basket Abandonment	6%	Under 2%
Staff Utilisation	70%	Above 90%
Average Queue Length	15 customers	Under 5 customers

These values are illustrative targets for the case study and demonstrate how project success could be measured.

7.6 Future State Summary

The proposed future state replaces a reactive checkout management process with a proactive, data-driven approach. By providing managers with real-time operational visibility and intelligent staffing recommendations, Tesco can improve customer experience, optimise workforce utilisation, and enhance overall operational performance while maintaining flexibility for human decision-making.

8. Gap Analysis

The purpose of the Gap Analysis is to compare the current ("As-Is") checkout process with the desired future ("To-Be") process. It identifies operational gaps that prevent Tesco from achieving its business objectives and recommends improvements to close those gaps.

8.1 Gap Analysis Table

Area	Current State (As-Is)	Future State (To-Be)	Gap Identified	Recommendation
Queue Monitoring	Queue lengths monitored manually by managers	Real-time monitoring through dashboard	No live operational visibility	Implement real-time queue monitoring dashboard
Staff Allocation	Staff assigned using fixed schedules	Staff allocated based on predicted demand	Staffing does not match customer demand	Introduce demand-based workforce planning
Decision Making	Decisions rely on manager experience	Decisions supported by operational data and AI recommendations	Limited data-driven decision making	Implement decision support tools
Checkout Management	Additional tills opened after queues become long	Additional tills recommended before queues exceed threshold	Reactive process	Introduce predictive alerts
Reporting	Limited manual reporting	Automated KPI dashboards and reports	Lack of operational visibility	Develop management dashboards
Customer Experience	Long queues during busy periods	Reduced waiting times	Customer dissatisfaction	Improve checkout efficiency
Performance Measurement	Few operational KPIs monitored	Real-time KPI monitoring	Poor performance visibility	Introduce KPI tracking
Workforce Utilisation	Staff sometimes under- or over-utilised	Dynamic staff allocation	Inefficient workforce planning	Optimise scheduling using operational data

8.2 Key Business Gaps

The analysis identified the following key gaps within the existing checkout process:

Gap 1 – Reactive Queue Management

Managers currently respond to long queues after they occur rather than preventing them through proactive monitoring.

Business Impact

- Longer customer waiting times
- Increased customer complaints
- Reduced operational efficiency

Gap 2 – Inefficient Staff Allocation

Current staffing schedules do not consistently reflect customer demand during peak trading hours.

Business Impact

- Understaffed checkouts
- Overworked colleagues
- Reduced service quality

Gap 3 – Limited Operational Visibility

Managers have limited access to real-time operational data to support decision-making.

Business Impact

- Delayed responses
- Inconsistent decisions
- Reduced productivity

Gap 4 – Limited Use of Data

Although operational and historical sales data exist, they are not fully utilised to improve workforce planning.

Business Impact

- Missed opportunities for optimisation
- Inefficient resource allocation

8.3 Gap Prioritisation

Gap	Priority	Business Impact
Reactive queue management	High	High
Inefficient staff allocation	High	High
Limited operational visibility	High	Medium
Limited use of operational data	Medium	Medium
Manual reporting	Medium	Low

8.4 Recommendations

To address the identified gaps, the following recommendations are proposed:

- Introduce real-time queue monitoring to improve operational visibility.
- Use predictive analytics to forecast customer demand and support workforce planning.
- Provide managers with AI-assisted staffing recommendations while retaining human decision-making.
- Implement operational dashboards displaying live KPIs and queue performance.
- Standardise decision-making using predefined operational thresholds and automated alerts.

8.5 Gap Analysis Summary

The analysis demonstrates that the current checkout process is primarily reactive and dependent on manual observation. The proposed future state introduces a proactive, data-driven approach to improve operational visibility, workforce planning, and the customer experience. Closing these gaps will help Tesco achieve its objectives of reducing waiting times, improving customer satisfaction, and increasing operational efficiency.

9. Business Requirements

9.1 Purpose

The purpose of this section is to define the business requirements necessary to improve Tesco's checkout operations. These requirements describe what the business needs to achieve rather than how the technical solution will be implemented.

9.2 Business Requirements Catalogue with Requirement Prioritisation (MoSCoW)

Requirement ID	Priority	Reason
BR-001	Must Have	Directly addresses the core business problem.
BR-002	Must Have	Essential for efficient operations.
BR-003	Must Have	Required for informed decision-making.
BR-004	Must Have	Supports revenue protection.
BR-005	Must Have	Key business objective.
BR-006	Should Have	Improves decision quality.
BR-007	Should Have	Supports operational monitoring.
BR-008	Should Have	Improves long-term workforce planning.
BR-009	Could Have	Improves management efficiency.
BR-010	Could Have	Valuable but not critical for initial release.

9.4 Business Rules

Business rules define policies the business expects the solution to follow.

Rule ID	Business Rule
BRR-001	The system shall generate an alert when average queue waiting time exceeds 5 minutes.

BRR-002	Only authorised managers may approve staffing recommendations.
BRR-003	Historical operational data shall be retained for a minimum of 12 months.
BRR-004	Operational dashboards shall refresh automatically at regular intervals.
BRR-005	All operational data shall comply with UK GDPR requirements.

9.5 Requirement Traceability

Each requirement should support at least one business objective.

Business Objective	Supporting Requirements
Reduce waiting times	BR-001, BR-002, BR-003
Improve customer satisfaction	BR-001, BR-004, BR-005
Improve operational efficiency	BR-002, BR-006, BR-007
Support data-driven decisions	BR-003, BR-006, BR-010

9.6 Business Requirements Summary

The business requirements focus on improving Tesco's checkout operations by reducing customer waiting times, optimising staff allocation, increasing operational visibility, and supporting proactive decision-making. These requirements provide the foundation for defining the functional and non-functional system requirements that follow.

10. Functional Requirements

10.1 Purpose

This section defines the functional requirements for the Tesco Smart Checkout Optimisation System. Functional requirements describe what the system must do to meet the business requirements.

10.2 Functional Requirements Catalogue

ID	Functional Requirement	Priority	Related Business Requirement
FR-001	The system shall monitor checkout queue lengths in real time.	Must Have	BR-001
FR-002	The system shall calculate the average customer waiting time for each checkout.	Must Have	BR-001
FR-003	The system shall generate an alert when the average waiting time exceeds 5 minutes.	Must Have	BR-001, BR-003
FR-004	The system shall recommend opening additional checkouts when demand exceeds predefined thresholds.	Must Have	BR-002
FR-005	The system shall recommend staff reallocation based on customer demand.	Must Have	BR-002
FR-006	The system shall provide Store Managers with a live operational dashboard.	Must Have	BR-003
FR-007	The dashboard shall display queue length, waiting time, active checkouts, staff allocation and customer traffic.	Must Have	BR-003
FR-008	The system shall generate daily, weekly and monthly operational reports.	Should Have	BR-010
FR-009	The system shall maintain historical operational data for trend analysis.	Should Have	BR-008

FR-010	The system shall allow Store Managers to manually override AI recommendations when required.	Must Have	BR-006
FR-011	The system shall log all recommendations and management decisions for audit purposes.	Should Have	BR-007
FR-012	The system shall display key operational KPIs in real time.	Must Have	BR-007

10.3 Functional Modules

Module 1 – Queue Monitoring

Purpose

Monitor customer queues continuously.

Functions

- Measure queue length.
- Calculate waiting time.
- Detect congestion.
- Send alerts.

Module 2 – Staff Allocation

Purpose

Support efficient workforce management.

Functions

- Recommend staffing changes.
- Identify available colleagues.
- Suggest checkout openings.
- Support manager approval.

Module 3 – Dashboard

Purpose

Provide operational visibility.

Functions

- Display live KPIs.

- Show queue performance.
- Show staffing levels.
- Display AI recommendations.

Module 4 – Reporting

Purpose

Support operational analysis.

Functions

- Daily reports.
- Weekly reports.
- Monthly reports.
- Trend analysis.

10.4 Functional Requirement Prioritisation

Priority	Description
Must Have	Required for the Minimum Viable Product (MVP).
Should Have	Important but not essential for the initial release.
Could Have	Additional features for future releases.

10.5 Functional Requirement Assumptions

The following assumptions apply:

- Queue data is available in real time.
- Staff availability data is maintained accurately.
- Managers have access to the dashboard during trading hours.
- Historical sales and customer traffic data are available.
- The AI recommendation engine integrates successfully with operational systems.

10.6 Functional Requirements Summary

The functional requirements define the key capabilities required to support smarter checkout operations. They provide real-time monitoring, intelligent staffing recommendations, operational reporting, and dashboard functionality to help store managers make informed decisions and improve customer service.

11. Non-Functional Requirements

11.1 Purpose

This section defines the quality attributes and operational characteristics of the Tesco Smart Checkout Optimisation System. These requirements ensure the solution is secure, reliable, performant, scalable, and usable while meeting Tesco's business and regulatory standards.

11.2 Non-Functional Requirements Catalogue

ID	Requirement	Category	Priority
NFR-001	The dashboard shall load within 2 seconds under normal operating conditions.	Performance	Must Have
NFR-002	The system shall process queue updates within 5 seconds of receiving new data.	Performance	Must Have
NFR-003	The solution shall achieve 99.9% availability during store operating hours.	Availability	Must Have
NFR-004	The system shall support multiple Tesco stores without significant performance degradation.	Scalability	Should Have
NFR-005	Only authorised users shall access management dashboards.	Security	Must Have
NFR-006	The system shall comply with UK GDPR requirements for data protection.	Compliance	Must Have
NFR-007	All user activity and management decisions shall be logged for audit purposes.	Auditability	Should Have
NFR-008	The interface shall be intuitive and require minimal staff training.	Usability	Must Have
NFR-009	The dashboard shall be accessible on desktop and tablet devices used by Store Managers.	Compatibility	Should Have
NFR-010	The solution shall recover from unexpected failures with minimal disruption to operations.	Reliability	Should Have

11.3 Performance Requirements

The system should:

- Display live queue information within **5 seconds** of changes.
- Refresh dashboard data automatically.
- Support multiple users simultaneously without noticeable delays.
- Generate reports in under **30 seconds**.

11.4 Security Requirements

The solution should:

- Require secure user authentication.
- Restrict access based on user roles (e.g., Store Manager, Shift Leader).
- Encrypt sensitive operational data during transmission.
- Record all critical user actions for audit purposes.

11.5 Availability & Reliability

The system should:

- Be available throughout normal store operating hours.
- Continue operating during temporary network interruptions where possible.
- Recover quickly after failures.
- Minimise service disruptions during planned maintenance.

11.6 Usability Requirements

The dashboard should:

- Present key operational KPIs clearly.
- Require minimal training for managers.
- Use consistent layouts and terminology.
- Allow users to find important information quickly.

11.7 Accessibility Requirements

The solution should:

- Use readable fonts and colour contrast.
- Be navigable using a keyboard where appropriate.
- Support accessibility standards adopted by Tesco.
- Present information in a clear, consistent format.

11.8 Compliance Requirements

The solution should:

- Comply with UK GDPR.

- Follow Tesco's internal information security policies.
- Retain operational data according to company retention policies.
- Ensure appropriate access controls for sensitive information.

11.9 Maintainability Requirements

The solution should:

- Allow updates with minimal downtime.
- Support future enhancements without major redesign.
- Use modular components where practical.
- Provide sufficient documentation for support teams.

11.10 Non-Functional Requirements Summary

The non-functional requirements ensure that the proposed solution is not only functional but also secure, reliable, scalable, user-friendly, and compliant with relevant regulations. These characteristics are essential for successful adoption and long-term operational effectiveness.

12. User Stories & Acceptance Criteria

12.1 Purpose

This section defines the user stories and acceptance criteria for the Tesco Smart Checkout Optimisation System. User stories capture the needs of different stakeholders from their perspective, while acceptance criteria define the conditions that must be met before a user story

12.2 User Story Catalogue

US-001 – View Real-Time Queue Information

User Story

As a Store Manager, I want to view real-time queue lengths and waiting times so that I can make informed operational decisions before queues become excessive.

Acceptance Criteria

- Queue lengths are displayed in real time.
- Average waiting time is shown for each checkout.
- Dashboard refreshes automatically.
- Queue status is colour-coded (Green, Amber, Red).

US-002 – Receive Queue Alerts

User Story

As a Shift Leader, I want to receive alerts when queue waiting times exceed the acceptable threshold so that I can open additional checkouts promptly.

Acceptance Criteria

- Alert generated when waiting time exceeds 5 minutes.
- Alert includes affected checkout.
- Alert remains visible until acknowledged.
- Alert time is logged.

US-003 – AI Staffing Recommendation

User Story

As a Store Manager, I want the system to recommend staff allocation based on customer demand so that checkout resources are used efficiently.

Acceptance Criteria

- Recommendation is generated using operational data.
- Manager can approve or reject the recommendation.
- Recommendation is recorded in the audit log.
- Dashboard displays recommendation status.

US-004 – Operational Dashboard

User Story

As a Store Manager, I want to view key operational KPIs in one dashboard so that I can monitor store performance efficiently.

Acceptance Criteria

Dashboard displays:

- Queue Length
- Waiting Time
- Active Checkouts
- Staff Allocation
- Customer Traffic
- AI Recommendations

US-005 – Operational Reports

User Story

As a Regional Operations Manager, I want to generate performance reports so that I can compare checkout efficiency across stores.

Acceptance Criteria

- Reports can be filtered by date.
- Reports can be exported.
- KPIs are included.
- Historical trends are displayed.

12.3 User Story Summary

ID	User	Goal	Priority	Story Points
US-001	Store Manager	Monitor queues	Must	8
US-002	Shift Leader	Receive alerts	Must	5
US-003	Store Manager	Receive recommendations staffing	Must	13
US-004	Store Manager	View dashboard	Must	8
US-005	Regional Operations Manager	Generate reports	Should	5

12.4 Business Value

These user stories ensure that the solution delivers measurable value by:

- Reducing customer waiting times.
- Improving operational visibility.
- Supporting proactive workforce management.
- Enhancing customer satisfaction.
- Enabling data-driven decision-making.

13. Product Backlog & MoSCoW Prioritisation

13.1 Purpose

The Product Backlog is an ordered list of features, enhancements, and requirements that will be delivered as part of the Tesco Smart Checkout Optimisation System. It helps ensure the project focuses on delivering the highest business value first.

The backlog has been prioritised using the **MoSCoW prioritisation technique**, categorising items as:

- **Must Have** – Essential for the solution to function.
- **Should Have** – Important but not critical for the first release.
- **Could Have** – Desirable enhancements if time and resources permit.
- **Won't Have (This Release)** – Excluded from the initial implementation but may be considered in future phases.

13.2 Product Backlog

Backlog ID	Epic	User Story	Priority	Story Points	Sprint
PB-001	Queue Monitoring	US-001 – View real-time queue information	Must Have	8	Sprint 1
PB-002	Alerts	US-002 – Receive queue alerts	Must Have	5	Sprint 1
PB-003	AI Staffing	US-003 – AI staffing recommendations	Must Have	13	Sprint 2
PB-004	Dashboard	US-004 – Operational dashboard	Must Have	8	Sprint 3
PB-005	Reporting	US-005 – Generate operational reports	Should Have	5	Sprint 4
PB-006	KPI Analytics	Performance trend analysis	Should Have	8	Sprint 4
PB-007	Notifications	Email and mobile alerts	Could Have	5	Future Sprint

PB-008	Customer Queue Display	Customer-facing queue information screens	Could Have	8	Future Sprint
PB-009	Multi-store Dashboard	Regional performance dashboard	Could Have	13	Future Sprint
PB-010	Automatic Workforce Scheduling	Fully automated staff scheduling	Won't Have (Phase 1)	21	Future Release

13.3 Epic Breakdown

Epic 1 – Queue Monitoring

Objective

Provide real-time visibility of queue performance.

Included Features

- Live queue monitoring
- Waiting time calculation
- Queue status indicators

Epic 2 – AI Decision Support

Objective

Support managers with intelligent staffing recommendations.

Included Features

- Demand forecasting
- AI recommendations
- Staffing optimisation

Epic 3 – Dashboard & Reporting

Objective

Provide operational visibility.

Included Features

- Live dashboard
- KPI monitoring
- Performance reports

Epic 4 – Future Enhancements

Objective

Support long-term operational improvements.

Included Features

- Mobile notifications
- Multi-store reporting
- Automated scheduling

13.4 Backlog Prioritisation Rationale

Must Have

These features directly address the core business problem of reducing checkout waiting times and improving operational efficiency.

Should Have

These features provide additional operational insights and reporting capabilities but are not essential for the first release.

Could Have

These enhancements improve usability and future scalability but do not prevent the system from delivering business value.

Won't Have (This Release)

These features require significant investment or organisational change and have therefore been excluded from the initial implementation.

13.5 MVP (Minimum Viable Product)

The MVP for Phase 1 will include:

- Real-time queue monitoring
- Waiting time alerts
- AI-assisted staffing recommendations
- Store Manager dashboard
- Basic operational reporting

This provides the highest business value while keeping the initial implementation manageable.

13.6 Product Backlog Summary

The backlog has been prioritised to ensure that the most valuable functionality is delivered first. By focusing on the MVP, Tesco can achieve measurable improvements in checkout performance while allowing additional features to be introduced in future releases based on business needs and stakeholder feedback.

14. Requirements Traceability Matrix (RTM)

14.1 Purpose

The Requirements Traceability Matrix (RTM) ensures that every business objective is translated into business requirements, functional requirements, user stories, and test cases. It helps verify that all business needs are addressed throughout the project lifecycle and supports change management, testing, and project governance.

14.2 Requirements Traceability Matrix

Business Objective	Business Requirement	Functional Requirement	User Story	Acceptance Criteria	Test Case
BO-001 Reduce customer waiting time	BR-001	FR-001 Monitor queue length	US-001 View real-time queue information	Queue length displayed in real time	TC-001 Verify live queue updates
BO-001 Reduce customer waiting time	BR-001	FR-002 Calculate waiting time	US-001	Waiting time displayed accurately	TC-002 Verify waiting time calculation
BO-002 Optimise staff allocation	BR-002	FR-004 Recommend additional checkouts	US-003 AI staffing recommendation	Recommendation generated when threshold exceeded	TC-003 Verify staffing recommendation

BO-002 Optimise staff allocation	BR-002	FR-005 Recommend staff allocation	US-003	Manager can approve or reject recommendation	TC-004 Verify manager approval
BO-003 Improve operational visibility	BR-003	FR-006 Dashboard	US-004 Operational dashboard	Dashboard displays KPIs	TC-005 Verify dashboard display
BO-004 Reduce basket abandonment	BR-004	FR-003 Queue alerts	US-002 Receive queue alerts	Alert generated after 5 minutes	TC-006 Verify queue alert
BO-005 Improve reporting	BR-010	FR-008 Generate reports	US-005 Operational reports	Reports export successfully	TC-007 Verify report generation

14.3 Traceability Summary

The matrix demonstrates that:

- Every **Business Objective** is supported by one or more **Business Requirements**.
- Every **Business Requirement** is implemented through **Functional Requirements**.
- Every **Functional Requirement** is represented by one or more **User Stories**.
- Every **User Story** has clear **Acceptance Criteria**.
- Every **Acceptance Criterion** can be validated through a **Test Case**.

This ensures complete end-to-end traceability from business need to system validation.

14.4 Benefits of Traceability

Maintaining a traceability matrix provides several advantages:

- Ensures all business objectives are addressed.
- Reduces the risk of missing or duplicated requirements.
- Simplifies impact analysis when requirements change.
- Supports testing and quality assurance.
- Improves communication between business and technical teams.
- Demonstrates governance throughout the project lifecycle.

14.5 Change Impact Example

Scenario: The business decides to reduce the maximum acceptable queue waiting time from **5 minutes to 3 minutes**.

The RTM immediately identifies the affected artefacts:

Artefact	Impact
Business Requirement BR-001	Update required
Functional Requirement FR-003	Update queue alert threshold
User Story US-002	Acceptance criteria updated
Test Case TC-006	Expected result updated
Dashboard KPI	Threshold updated

15. Solution Design

15.1 Purpose

The purpose of this section is to describe the proposed solution that addresses the identified business problems and fulfils the documented business and functional requirements. The solution is designed to improve checkout efficiency, enhance customer experience, and support data-driven operational decision-making while maintaining managerial oversight.

15.2 Proposed Solution Overview

The proposed solution is a **Smart Checkout Optimisation System** that combines real-time queue monitoring, predictive analytics, and operational dashboards to assist Store Managers and Shift Leaders in managing checkout operations.

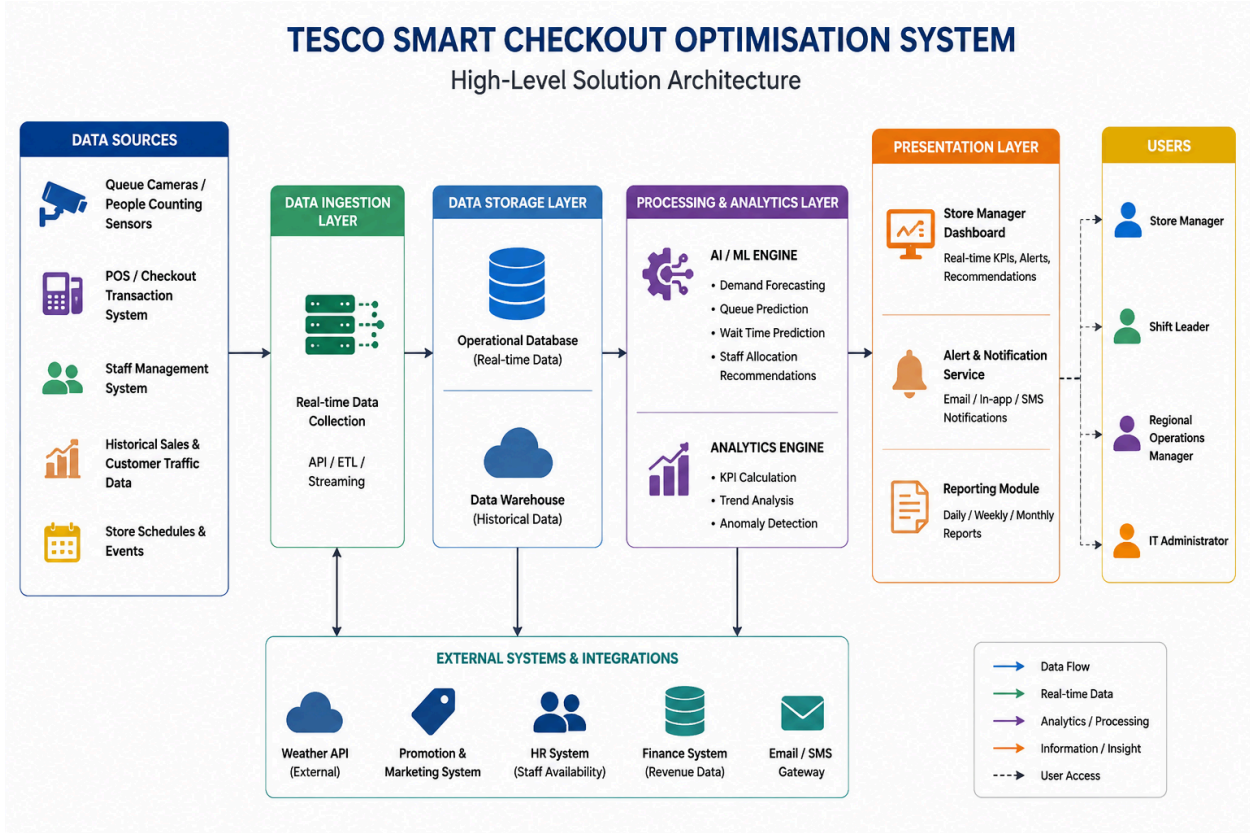
The solution does **not** replace human decision-making. Instead, it provides intelligent recommendations that enable managers to make faster and better-informed operational decisions.

15.3 Solution Components

Component	Description
-----------	-------------

Queue Monitoring Module	Tracks queue lengths and customer waiting times in real time.
AI Demand Forecasting Engine	Predicts customer traffic using historical and operational data.
Staff Allocation Module	Recommends staff deployment based on predicted demand.
Operational Dashboard	Displays KPIs, alerts, queue status and staffing recommendations.
Reporting Module	Generates daily, weekly and monthly performance reports.
Alert & Notification Service	Sends notifications when predefined thresholds are exceeded.

15.4 High-Level Solution Architecture



15.5 User Roles

User Role	Responsibilities
Customer	Purchase products through checkout.
Checkout Colleague	Process customer transactions.
Shift Leader	Monitor queues and manage staffing.
Store Manager	Review dashboards, approve recommendations and monitor KPIs.
Regional Operations Manager	Review performance across multiple stores.
IT Administrator	Maintain the solution and manage system access.

15.6 Business Benefits

The proposed solution is expected to deliver the following benefits:

Customer Benefits

- Reduced waiting times.
- Improved shopping experience.
- Faster checkout process.
- Increased customer satisfaction.

Operational Benefits

- Better workforce utilisation.
- Improved visibility of checkout operations.
- Data-driven staffing decisions.
- Standardised operational processes.

Financial Benefits

- Reduced basket abandonment.
- Increased completed sales.
- Improved operational efficiency.
- Better return on staffing investment.

15.7 Success Metrics

KPI	Current	Target
Average Waiting Time	8 min	< 3 min
Customer Satisfaction	78%	> 90%
Basket Abandonment	6%	< 2%
Staff Utilisation	70%	> 90%
Manager Response Time	8 min	< 2 min

15.8 Solution Limitations

The proposed solution has the following limitations:

- AI recommendations depend on the quality of available operational data.
- Human approval remains necessary for staffing decisions.
- Forecast accuracy may vary during exceptional trading periods (e.g., holidays or special events).
- Successful implementation depends on user adoption and staff training.